Operating Systems Project: Process Scheduling Simulation

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1. Overview

This project simulates CPU process scheduling using different algorithms.

Implemented Algorithms:

- First-Come, First-Served (FCFS) – Processes run in order of arrival.
- Shortest Job First (SJF) – Processes with smallest burst time run first.
- Round Robin (RR) – Each process receives a fixed time quantum in rotation.
- Memory Management (Bonus) – First-Fit allocation of memory to processes.

The program reads process information from an input file (processes.txt) and displays a text-based Gantt chart, waiting time (WT), turnaround time (TAT), and CPU utilization.

2. Implementation Details

2.1 Input File

The program uses a text file processes.txt with the following structure:

PID	Arrival Time	Burst Time	Priority	Memory
1	0	5	2	20
2	2	3	1	10
3	4	2	3	30
4	5	4	2	25
5	6	1	4	15

- **Language:** C++
- **Input:** processes.txt
- **Output:** Gantt chart, process WT/TAT, CPU utilization, memory allocation table
- **Memory Management:** First-Fit allocation for each process
- **Scheduling Algorithms Implemented:** FCFS, SJF (Non-Preemptive), Round Robin
- **Data Structures:** struct Process to store PID, arrival time, burst time, priority, memory, waiting time, and turnaround time. Vector<Process> to store all processes.

2.2 Scheduling Algorithms

1. FCFS

- Sort processes by arrival time.
- Calculate WT = start time – arrival time, TAT = WT + burst time.
- Gantt chart shows execution order.

2. SJF (Non-Preemptive)

- Select the available process with the shortest burst time at each step.
- Accounts for waiting if the CPU is idle.

3. Round Robin

- Each process executes for a fixed time quantum.
- Remaining burst time is tracked for next turns.
- Handles idle time if no process is ready.

Memory Management (Bonus)

- Implemented First-Fit Allocation.
- Each process is assigned a memory block if available and has start and end addresses.

3. Sample Run

Screenshot of program running FCFS:

```
mina@HP:~/OS_Project$ ./scheduler | tee output_FCFS.txt
First-Fit Memory Allocation:

Memory Allocation (First-Fit):
Process MemoryReq       StartAddr       EndAddr
P1      20              0               19
P2      10              20              29
P3      30              30              59
P4      25              60              84
P5      15              85              99

Select Scheduling Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Round Robin
Choice: 1

Gantt Chart:
| P1 | P2 | P3 | P4 | P5 |
0 5 8 10 14 15

Process WT      TAT
P1      0       5
P2      3       6
P3      4       6
P4      5       9
P5      8       9
Average WT: 4.00
Average TAT: 7.00
CPU Utilization: 100.00%
```

Screenshot of program running SJF:

```
mina@HP:~/OS_Project$ ./scheduler | tee output_SJF.txt
First-Fit Memory Allocation:

Memory Allocation (First-Fit):
Process MemoryReq       StartAddr       EndAddr
P1      20              0               19
P2      10              20              29
P3      30              30              59
P4      25              60              84
P5      15              85              99

Select Scheduling Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Round Robin
Choice: 2

Gantt Chart:
| P1 | P3 | P5 | P2 | P4 |
0 5 7 8 11 15

Process WT      TAT
P1      0       5
P2      6       9
P3      1       3
P4      6       10
P5      1       2
Average WT: 2.80
Average TAT: 5.80
CPU Utilization: 100.00%
```

Screenshot of program running RR:

```
mina@HP:~/OS_Project$ ./scheduler | tee output_RR.txt
First-Fit Memory Allocation:

Memory Allocation (First-Fit):
Process MemoryReq       StartAddr       EndAddr
P1      20              0               19
P2      10              20              29
P3      30              30              59
P4      25              60              84
P5      15              85              99

Select Scheduling Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Round Robin
Choice: 3
Enter time quantum: 2

Gantt Chart:
| P1 | P2 | P3 | P4 | P5 | P1 | P2 | P4 | P1 |
0 2 4 6 8 9 11 12 14 15

Process WT      TAT
P1      10      15
P2      7       10
P3      0       2
P4      5       9
P5      2       3
Average WT: 4.80
Average TAT: 7.80
CPU Utilization: 100.00%
```

4. Challenges & Solutions

Challenge	Solution
Handling idle CPU when no process is ready	Added check to increment time until a process arrives
Displaying Gantt chart neatly	Stored start times and process IDs in vectors for printing
Round Robin time slicing	Tracked remaining burst time for each process
Memory allocation	Implemented First-Fit algorithm and displayed start/end addresses

5. Conclusion

This project successfully demonstrates the simulation of process scheduling using FCFS, SJF, and Round Robin algorithms, along with memory management through First-Fit allocation. The program effectively calculates waiting time, turnaround time, and CPU utilization, and presents results via Gantt charts and output files.

Overall, the project illustrates how an operating system manages CPU scheduling and memory allocation in a simplified but realistic manner, providing both functional understanding and practical application of key OS concepts.